Muhammad Usman

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Personal Information

Age: 33 years **Nationality:** Pakistani

Research Objective

Ph.D. candidate in Information and Communication Engineering at Shenzhen University (expected November 2026), specializing in self-supervised 3D human-motion representation learning. My research combines contrastive learning, geometric algebra, graph-based models, spatiotemporal Transformers, and knowledge distillation. I seek a postdoctoral position to advance geometry-aware and efficient learning for human-motion understanding, multimodal perception, and intelligent systems.

Research Interests

Self-Supervised and Contrastive Learning; 3D Skeleton-Based Action Recognition; Geometric Algebra and Geometric Deep Learning; Graph Neural Networks and Spatiotemporal Transformers; Knowledge Distillation; Multimodal Human-Motion Understanding.

Education

2022 – present **Ph.D., Information and Communication Engineering**, Shenzhen University, China.
Expected completion: November 2026.
Thesis: *Research on Key Technologies of Contrastive Learning for Unsupervised 3D Skeleton-Based Action Recognition*.

2018 – 2020 **M.Eng., Electrical and Computer Engineering**, South China University of Technology, China.
Thesis: *Research on the Method of Judging Litchi Maturity & Size Based on Image Deep Learning*.

2012 – 2016 **B.Sc., Telecommunication and Networking**, COMSATS University Islamabad (Abbottabad Campus), Pakistan.
Thesis: *Agent Negotiation (Java)*.

Research Experience

2022 – present **Doctoral Researcher**, Shenzhen University, China.
Developed self-supervised skeleton action-recognition methods using hierarchical contrastive learning and knowledge distillation; investigated geometric-algebra-guided and graph/Transformer-based spatiotemporal models.
Evaluated representation quality and model efficiency on established 3D skeleton action-recognition benchmarks.

Teaching and Professional Experience

2017 – 2018 **Lecturer**, Northern University, Nowshera, Pakistan.

2016 – 2017 **Intern**, Pakistan Telecommunication Company Limited (PTCL), Pakistan.

Research Publications

Peer-Reviewed Journal Articles

1. Cao, W. (Supervisor as first author), M. **Usman**, and X. Yin, "Hierarchical joint contrastive learning with knowledge distillation for self-supervised 3d skeleton-based action recognition," *Expert Systems with Applications*, 2026, Art. no. 131204.
2. M. **Usman**, W. Cao, Z. Huang, J. Zhong, and R. Ji, "Otm-hc: Enhanced skeleton-based action representation via one-to-many hierarchical contrastive learning," *AI*, vol. 5, no. 4, pp. 2170–2186, 2024.
3. M. **Usman** and J. Zhong, "Skeleton-based motion prediction: A survey," *Frontiers in Computational Neuroscience*, vol. 16, 2022, Art. no. 1051222.
4. F. Banoori, J. Shi, J. Arshad, R. Han, and M. **Usman**, "Peak to average power ratio alleviation by utilizing swarm intelligence along with machine learning for multiple input multiple output-orthogonal frequency division multiplexing based underwater acoustic communication system," *Transactions on Emerging Telecommunications Technologies*, vol. 32, no. 11, e4335, 2021.

Manuscripts Under Review

1. M. **Usman** and W. Cao, "Ga-scl: Geometry-aware self-supervised contrastive learning with geometric algebra transformers for 3d skeleton action recognition," 2026, Manuscript under review at Expert Systems with Applications (Elsevier).
2. M. **Usman**, W. Cao, and X. Yin, "Geometric algebra-based contrastive learning for frequency and language-aware skeleton action recognition," 2026, Manuscript under review at Pattern Recognition (Elsevier).

Conference Proceedings

1. S. Li, S. Waleed, and M. **Usman**, "Real time object tracking system using ultraviolet laser," in *2020 Chinese Control and Decision Conference (CCDC)*, IEEE, 2020, pp. 3003–3008.

Technical Skills

Programming & Tools	Python, C++, MATLAB, Java; PyTorch, Linux, GPU-Based Model Training, LaTeX.
Machine Learning	Self-Supervised and Contrastive Learning, Knowledge Distillation, Representation Learning, Experimental Design and Model Evaluation.
Computer Vision	3D Skeleton-Based Action Recognition, Human-Motion Analysis, Spatiotemporal Feature Learning, Intelligent Visual Perception.
Geometric & Graph AI	Geometric Algebra, Geometric Deep Learning, Graph Neural Networks, Graph Convolutional Networks, Graph/Spatiotemporal Transformers.
Languages	English (fluent); Mandarin Chinese (intermediate).

Honors and Awards

2018 **Chinese Government Scholarship (CGS), Silk Road Program.** For master's study at South China University of Technology.

2022 **Guangdong Government Outstanding International Student Scholarship and Shenzhen University Scholarship.** Four-year Ph.D. scholarship support.

2024 **Guangdong Government Scholarship (RMB 30,000).** For published research achievements.

2025 **Shenzhen University Outstanding Innovative Talent Scholarship (Second Prize; RMB 17,000).** For research achievements and innovation.

Academic References

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